Ikrar Gempur Tirani

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Summary

Final-year Informatics Engineering student at Hasanuddin University (GPA 3.91/4.00) specializing in Artificial Intelligence, with hands-on experience building production LLM and Retrieval-Augmented Generation (RAG) systems. Currently an AI Engineer intern on the UniAI institutional chatbot team, working across the full ML lifecycle: data pipelines, embedding and retrieval, model serving with vLLM, and deployment. Experienced in natural language processing and applied machine learning, from fine-tuning transformer models to statistical modeling, complemented by full-stack engineering skills and leadership experience heading a creative media team at Google Developer Groups on Campus.

Education

Hasanuddin University — Makassar, South Sulawesi
Bachelor of Informatics Engineering (AI Specialization) – GPA: 3.91/4.00 — *Aug. 2023 – Present*

MAN Insan Cendekia Gorontalo — Gorontalo
Science — *Aug. 2020 – June 2023*

Technical Skills

Programming Languages: Python, SQL, JavaScript, TypeScript, Java, C++, Dart
Machine Learning & NLP: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NLTK, Fine-tuning, Statistical Modeling
LLM & RAG Engineering: vLLM, LlamaIndex, Qdrant (Vector Database), RAG Pipelines, Embedding Models, Reranking, Prompt Engineering, Semantic Caching (Redis), Llama Guard, OCR (Tesseract, PyMuPDF)
Data Analysis & Visualization: Pandas, NumPy, Polars, Jupyter Notebook, Matplotlib, Seaborn, Exploratory Data Analysis
Backend & Databases: FastAPI, Flask, Node.js, Express.js, PostgreSQL, MySQL, Redis, Supabase, Firebase, Nginx
Web & App Development: Next.js, React, Flutter, HTML/CSS
Tools & Platforms: Git, GitHub, Docker, Linux, Google Colab, VS Code, Vercel, Railway, VPS Deployment
Languages: Bahasa Indonesia (Native), English (Upper-Intermediate), Arabic (Intermediate)

Experience

AI Engineer Intern – RAG Core (UniAI Team) — Feb. 2026 – Present
Universitas Hasanuddin (DSITD) — *Makassar, South Sulawesi*

- Build the data and RAG core of UniAI, an institutional academic chatbot for Hasanuddin University, serving students and staff with university-specific knowledge; part of Team 2 (Data & RAG Core) within a three-team AI engineering organization.
- Developed end-to-end RAG pipeline using Qwen3-Embedding-0.6B for dense retrieval, BGE-Reranker for relevance reranking, Qdrant vector database, and Redis semantic caching to reduce latency and inference cost on repeated queries.
- Designed a 6-layer safety and routing pipeline (keyword filter, Llama Guard moderation, IndoBERT intent classifier, private API handler, RAG pipeline, output filter) to ensure safe and accurate responses for institutional use.
- Engineered a three-tier document OCR ingestion strategy (PyMuPDF, Tesseract, Qwen3-VL vision model fallback) to convert heterogeneous academic PDFs into high-quality retrieval corpora.
- Deployed and served Qwen3-VL-8B-Instruct via vLLM behind FastAPI and Nginx on an NVIDIA L40S GPU, tuning vLLM serving parameters (GPU memory utilization, batching) and load-testing toward 500 concurrent users.

Full Stack Developer Intern — Feb. 2026 – Apr. 2026
PLN Icon Plus (Iconnet), SBU Sulawesi & IBT — *Makassar, South Sulawesi*

- Designed and built an internal OPEX monitoring dashboard end-to-end using Next.js (TypeScript) and PostgreSQL to streamline budget tracking and realization reporting for regional operations.
- Performed data reconciliation across financial and operational records, automating previously manual reporting workflows and improving data consistency for management reporting.
- Deployed the application to a Linux VPS and hardened it through a full penetration testing cycle, resolving all identified security vulnerabilities across authentication, access control, and input validation.

Head of Creative Media Division — Aug. 2025 – Present
Google Developer Group on Campus – Hasanuddin University — *Makassar, South Sulawesi*

- Lead a team of 6, growing the Instagram community by 46% (2,800 to 4,100 followers) and generating 1M+ total content views, with a record 515K monthly views (25x pre-leadership baseline).
- Analyze engagement metrics with Instagram Insights to drive data-informed decisions on content format, posting schedule, and creative direction; supported production of Build with AI Makassar 2026 event materials.

Fullstack Developer (Freelance) June 2025 – Jan. 2026
Cirebon Kuring Cafe *Remote*

- Developed web and mobile applications using Next.js (TypeScript) and Flutter with a PostgreSQL database covering cafe operations, inventory tracking, and customer ordering.
- Translated stakeholder business requirements into technical specifications and database schema design.

Publication, Design, and Documentation Coordinator Sep. 2024 – Apr. 2025
Recursion UH *Makassar, South Sulawesi*

- Coordinated publication strategy for an inaugural national-level informatics competition (CTF, UX Design, ICT Business Plan, Competitive Programming), building its social media presence from zero to 894 followers with 96+ published content pieces in 8 months.

Projects

Urban Complaint Pattern Mining on NYC 311 Data (21M+ Records) June 2026
Big Data and Data Mining Course

- **Tech Stack:** Python, Polars, mlxtend (FP-Growth), Scikit-learn, SciPy, Matplotlib
- Mined 21.3M NYC 311 service requests end-to-end with a Polars pipeline, transforming 18.85M cleaned records into 764,114 building-level transaction baskets and extracting 95,902 frequent itemsets via FP-Growth association rule mining.
- Discovered a six-complaint "interior maintenance syndrome" (mean pairwise lift 8.48, bootstrap 95% CI [7.44, 9.55]) and validated it against 11M independent housing inspection records: buildings with the full syndrome showed 12–44x higher verified violation rates, holding across building-size strata and a prior time window designed to rule out administrative circularity.
- Clustered 6 years of ZIP-level complaint compositions with K-means (silhouette-based model selection, 20-seed stability testing), revealing three persistent neighborhood typologies with 84% temporal stability and no significant correlation to census socio-economic shifts.

Financial News Aspect-Based Sentiment Analysis Sep. 2025 – Nov. 2025
Natural Language Processing Course

- **Tech Stack:** Python, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, CUDA
- Fine-tuned RoBERTa-base (124M parameters) for aspect-based sentiment analysis on a 14,446-sample financial news dataset, achieving 90.74% F1-macro and 90.64% accuracy for entity-level market sentiment tracking.
- Implemented anti-leakage data splitting and EDA revealing severe class imbalance (60% neutral), then applied weighted cross-entropy, label smoothing, dropout, L2 regularization, and early stopping for robust generalization.
- Optimized GPU training with AdamW, gradient clipping, and learning rate warmup, reaching best performance at epoch 6.

E-Commerce Trust Simulation with LLM Agents Sep. 2025 – Nov. 2025
Simulation and Modeling Course

- **Tech Stack:** Python, MESA (Agent-Based Modeling), Llama 3.1 8B, Ollama, Pandas, SciPy, Matplotlib
- Simulated an e-commerce marketplace with 7,500+ autonomous agents (1,200 genuine reviewers, 380 fake reviewers, 6,000 shoppers) across 20 iterations to quantify consumer vulnerability to coordinated fake review campaigns.
- Demonstrated fake review attacks raise conversion rates by 54–72 percentage points for low-quality products, validated with Chi-Square (X^2=121–177, $p < 0.0001$), ANOVA, and Cramer's V.
- Engineered 3 LLM-powered behavioral personas (Impulsive, Careful, Skeptical) with local Llama 3.1 8B, generating 1,580+ realistic reviews at zero cloud API cost, plus an automated statistical analysis and visualization pipeline.

Medical Anamnesis Chatbot with NLP (Chatbot PUSTU) Aug. 2025 – Dec. 2025
Natural Language Processing Course

- **Tech Stack:** Python, Scikit-learn, Flask, Next.js, Gemini Flash 2.0 API
- Built a full-stack medical chatbot with 92.61% intent classification accuracy (Multinomial Naive Bayes + TF-IDF, 5,000 features, 1–2 n-grams) to streamline patient anamnesis for Indonesian Puskesmas healthcare workers.
- Automated training data generation with Gemini Flash 2.0 and custom prompt engineering, producing 14,000 balanced samples across 14 intent classes with a validation pipeline for Indonesian medical terminology.
- Developed a custom Indonesian NLP preprocessing pipeline (slang normalization, stopword filtering) and dictionary-based NER covering 97 symptom types and 23 body locations; deployed on Railway and Vercel with PDF export and stateful 14-stage dialog management.

Achievements

Finalist – UI/UX Design Competition, Identitas Manajemen Informatika Politeknik Astra 2024